Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/23/2006
PI Name:	Connolly, Janis M.S.		
Project Title:	Wireless Crew Communication		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	9
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	01/01/2003	**End Date:**	09/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Whitmore, Mihriban (Habitabiltiy and Human Factors Office)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	Reports from completed ISS missions indicate that issues associated with communication require crew time and cause inefficiencies in operations. Interrupting tasks to travel to an audio terminal unit (ATU) to communicate with the ground or crew in other modules requires time and impacts efficiency. In addition, noise interferes with communication even within the same module. There is a need to reduce the operational overhead associated with communication on ISS. Wireless communication would address this need. The long range goal is to have a unified integrated wireless technology system that supports all wireless activities on ISS and has well-designed and tested human interface options to support the range of operations on ISS. The goal of this project is to define the functional, performance, and operational requirements for wireless communication system, and develop and test a prototype for wireless functionality as well as usability, human interface, and ergonomics. A wireless communication design team, including human factors as well as electrical engineers will determine the necessary requirements via industry review, task analysis, and focus		

Task Description:	groups, and will complete a series of iterations of design integrating human information and control needs with the constraints of the technology. Each prototype iteration will be evaluated with increasing degrees of human factors assessment ranging from review by human factors professionals to formal usability testing. This project will significantly advance the ISS toward a wireless communication system by defining conclusively the requirements for such a system and by providing a evaluated prototype. Wireless communication will save crew time by reducing the need to travel to other crewmembers and to ATU units, and it will reduce the impact of ground interruptions on ongoing tasks. Wireless technology also has the potential to reduce risk by enabling better communication capabilities during emergencies. A single unified wireless system could further make operations even more efficient by reducing overhead associated with switching between systems. Near term: provide wireless human system interface to International Space Station communication system. Long-term: Identify human factors and functional requirements for wireless communications and conceptual design of user interfaces. Provide wireless communications to existing space program to enhance crew efficiency and productivity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Specific earth benefit has not been identified at this point in the technical development process.
Task Progress:	Avionics Division has completed the hardware circuit board; checkout and testing in progress. Completed human factors requirements for the near and long-term implementations. Initial conceptual design work for human user interface is completed. Follow-on design iteration continues. Software development is underway.
Bibliography Type:	Description: (Last Updated: 01/26/2005)
Presentation	Susan Baggerman/Janis Connolly – NASA Johnson Space Center J. Dory, L. Hua, L. Webb, P. Sinnott Habitability & Environmental Factors Office E. Burrell, M. Cooke, R. Villareal, J. Gard – NASA Johnson Space Center Avionics Systems Division "Wireless Crew Communication System Concept Demonstration" Habitation 2004 Conference on Space Habitation Research and Technology Development Jan-2004